

Vision-Based Vibration Analysis

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Abstract:

Traditional vibration analysis methods for condition monitoring of rotating machinery rely heavily on physical sensors, which can be costly, intrusive, and difficult to deploy in certain industrial environments. This paper presents a novel vision-based vibration analysis framework that leverages standard video footage and deep learning techniques to perform non-contact fault detection. The proposed pipeline consists of five stages: data acquisition, Eulerian Video Magnification (EVM), keypoint motion analysis, time-series signal generation, and anomaly detection. EVM is applied to enhance subtle motion in video frames, enabling the extraction of fine-grained vibration patterns. These patterns are tracked using keypoint motion analysis and converted into one-dimensional signals that represent the dynamic behavior of machine components. A Long Short-Term Memory (LSTM) network is trained to extract temporal features from these sequences, and a Support Vector Machine (SVM) is subsequently used for classification. Experimental results demonstrate that the system achieves a classification accuracy of 97% on a binary fault detection task using an 80/20 train-test split. This study highlights the potential of vision-based systems as scalable, low-cost alternatives to traditional sensor-based monitoring techniques for rotating machinery.

1. Introduction

Rotating machinery requires continuous vibration monitoring to prevent failures and ensure safety. While traditional sensor-based methods (like accelerometers) provide accurate measurements, they face three key limitations: high costs, complex installation, and inability to monitor multiple points simultaneously. Vision-based alternatives offer a promising solution but struggle to detect sub-millimeter vibrations in standard video footage.

2. Proposed Method

Our computer vision pipeline overcomes these challenges through five key stages:

2.1 Motion Magnification

We employ Eulerian Video Magnification (EVM) to amplify subtle vibrations in raw video. This enhancement makes sub-pixel motions visible for analysis.

2.2 Frame Extraction

The magnified video is decomposed into individual frames at a fixed sampling rate

2.3 Feature Tracking

From each extracted frame, we isolate measurable features including mean intensity, edge sharpness, and motion, that capture vibration patterns. These features are then prepared for time-series analysis in the next processing stage.

2.4 Time-Series Conversion

Feature displacements are converted into vibration waveforms, creating a time-series dataset compatible with signal processing tools.

2.5 LSTM -Based Vibration Pattern Recognition

The processed time-series data undergoes deep learning analysis through a structured pipeline. The extracted frame features - mean intensity, edge sharpness, and motion magnitude - are standardized using z-score normalization to ensure consistent scaling across input dimensions, preventing any single feature from dominating the learning process.

The normalized features are structured into sequential windows of 10 timesteps, where each window contains both spatial characteristics (intensity and edge sharpness) representing vibration-induced visual changes and motion features capturing displacement magnitudes between frames. This sliding window approach preserves the essential temporal relationships for vibration analysis while generating sufficient training samples.

A 64-unit LSTM layer processes the input sequences (10 timesteps $\times$ 3 features) to learn long-range dependencies in vibration patterns, extract compact temporal representations, and output fixed-length feature vectors encoding the vibration signatures. The LSTM's hidden state outputs feed into a support vector machine classifier with radial basis function kernel for final anomaly detection.

2.6 Support Vector Machine for Anomaly Classification

After temporal features are extracted by the LSTM, we pass the resulting feature vectors into a Support Vector Machine (SVM) classifier with a radial basis function (RBF) kernel. The SVM is trained to distinguish between normal and faulty states based on the learned temporal patterns in the vibration signals. This combination leverages LSTM's sequence modeling strengths and the SVM's robustness in binary classification tasks, particularly in cases with limited and high-dimensional data.

The model achieves 97% classification accuracy when evaluated using an 80-20 train-test split, with performance metrics including confusion matrix analysis and per-class precision/recall/F1 scores. This hybrid architecture effectively combines the LSTM's temporal modeling strengths with SVM's robust classification capabilities for vibration analysis.

Conclusion:

This paper presents a vision-based vibration monitoring system using video magnification and LSTM neural networks. Our approach detects machinery vibrations with 97% accuracy from standard video, matching physical sensors while being more affordable. The LSTM effectively learns vibration patterns from magnified video frames, providing a practical solution for industrial monitoring. Future work will improve high-frequency detection and lighting robustness.

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